Compliant Mechanism for Precise Maneuvering of Endotracheal Intubation Robot

Kenny Tanaka
College of Engineering
The Ohio State University
Columbus, OH
tanaka.90@osu.edu

***Abstract*—The purpose of this project is to investigate the implementation of a compliant mechanism as a replacement for a segmented soft robotic chain for automated endotracheal intubation. The previous design utilized a series of rigid links actuated by tendons to achieve bending within the confined geometry of an endotracheal tube, which introduced challenges in force actuation and precise controllability. This work proposes an alternative solution by introducing a compliant mechanism with distal steering for smoother and more precise bending behavior. As part of this study, an analytical formulation will be derived using the pseudo-rigid body (PRB) method by approximating the beam behavior through a series of rigid links with torsional springs to relate tendon actuation with the tip deflection and curvature. Finite element analysis (FEA) will be used to validate the analytical predictions and evaluate stress and deformation characteristics. The results will assess the feasibility of compliant mechanisms for precision-guided intubation.**

I. Introduction

The Department of Biomedical Engineering at The Ohio State University had 22 capstone projects in the year 2025. Among these projects was an AI-assisted Endotracheal Intubation robot designed to guide and maneuver the endotracheal tube into a patient's trachea while adapting to physiological conditions in real time [1]. While this prototype demonstrated promising results, the team leading this project reported facing a variety of issues with the soft robot design that controlled and maneuvered the endotracheal tube, lacking precise control of its endpoint due to the several points of deflection created by each link. This limitation serves as the main motivation for this work by introducing a compliant mechanism, seeking to provide a solution that offers increased control and predictability through a single part without sacrificing the flexibility required for navigating the airway through the use of a compliant mechanism.

II. Scope

The project focuses on the accurate modeling and analysis of a compliant mechanism in an effort to replace the segmented soft robotic design of the endotracheal intubation robot. To achieve this, the mechanism will be modeled as a planar compliant beam approximated by a series of rigid links and torsional springs. The project will be based on Dr. Su's model and analysis of a pseudorigid-body 3R model [2] to guide the development of the model. Subsequently, the goal of this project is to vary key parameters such as the tendon force and moment applied, the material property, and the length and number of links, to better fit the application of the endotracheal intubation robot. The PRB model results will be validated with a finite element analysis model. Lastly, both models would be implemented into an interactive computer application with a graphical user interface (GUI) that allows for the variation of key design and loading parameters.

III. Problem Definition

The motion of the endotracheal intubation robot consists of a primary motion that provides the driving force for inserting the endotracheal tube into the patient's trachea, followed by small adjustments to guide the tube through the trachea while adapting to the patient's unique physiology.

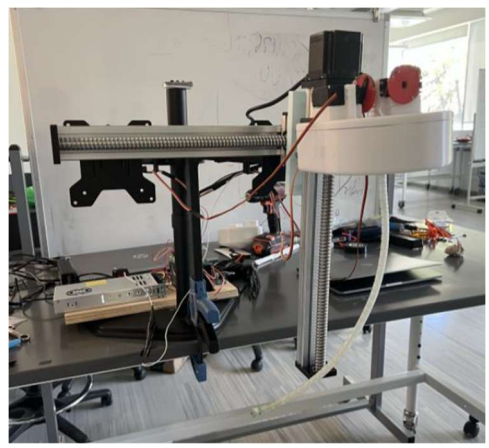

Fig. 1 Functional Prototype of Endotracheal Intubation Robot

The intubation tube has an outer diameter that fits securely within the patient's trachea and is designed to maintain its structural integrity within it to provide a stable pathway for air. As a result, the robot has to balance dimensional, motion tracking, and stiffness constraints, being able to fit within the tube while controlling it precisely and bending it from within. The pseudorigid-body formulation and finite element analysis formulation will be performed to assess the efficacy of a compliant mechanism against these requirements.

References

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